1.3 Right Triangle Trigonometry

Name _____ Period ____ Date _____

Pre-Calculus 1.3 · Student sheet

I can…

- evaluate the six trigonometric functions of an acute angle in a right triangle.
- use special angles and the fundamental identities.
- use trigonometry to find missing sides in real-life problems.

Vocabulary · fill in the missing word

Right triangle definitions — $\sin\theta =$ _____, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$

Cofunctions — $\sin\theta =$ _____, $\tan\theta = \cot(90° - \theta)$, $\sec\theta = \csc(90° - \theta)$

Pythagorean identity — $\sin^2\theta + \cos^2\theta =$ _____

Angle of elevation — the angle from the _____ up to the line of sight

Part 1 · Same angle, same ratio

Three right triangles, each with a 35° angle.

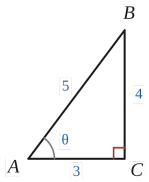
Triangle	Opposite	Hypotenuse	Opposite ÷ hypotenuse
Small			
Medium			
Large			

Part 2 · Notes and examples (fill in each note, then work the example)

SOH CAH TOA: $\sin\theta = \text{opp/hyp}$, $\cos\theta = \text{adj/hyp}$, $\tan\theta =$ _____.

Example 1

Use the triangle to evaluate all six trigonometric functions of θ.



Special angles: $\sin 30° = 1/2$, $\sin 45° = \sqrt{2}/2$, $\sin 60° =$ _____.

Example 2

Find the exact values.

a. $\sin 45°$

b. $\cos 60°$

c. $\tan 30°$

d. $\sin 30°$ and $\cos 60°$: what do you notice?

Identities: $\sin^2\theta + \cos^2\theta =$ _____; $\tan\theta = \sin\theta/\cos\theta$.

Example 3

θ is acute and $\sin\theta = 0.6$. Find $\cos\theta$ and $\tan\theta$.

Example 4

A surveyor stands 115 feet from the base of a monument. The angle of elevation to the top is 78.3°. How tall is the monument?

Example 5

θ is acute. Find θ in degrees and radians.

a. $\cos\theta = \frac{\sqrt{2}}{2}$

b. $\tan\theta = \sqrt{3}$

Part 3 · Partner practice

Partner 1

You stand 50 feet from a flagpole. The angle of elevation to the top is 40°. How tall is the flagpole?

Partner 2

θ is acute and $\cos\theta = \frac{5}{13}$. Find $\sin\theta$ and $\tan\theta$.